## Investigation: MUNDIAL S.A - PRODUTOS DE CONSUMO (CVM code 5312), fiscal year 2010, as it was on 2012-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

## Company

Sector: Têxtil e Vestuário (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

## Document used

2010 DFP, version 2, filed on 2011-04-04; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2011-04-04); 2 (2011-04-04).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=6315

## Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −2.106 (index)
- Alert by the study 1 rule: no
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: 75 (n = 22)
- Percentile in group and year: 73 (n = 347)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 1.077 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 83 (n = 26)
- Percentile in group and year: 69 (n = 457)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 1.005 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 63 (n = 26)
- Percentile in group and year: 63 (n = 467)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 1.046 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 48 (n = 26)
- Percentile in group and year: 67 (n = 471)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 1.269 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 75 (n = 26)

Percentile in group and year: 71 (n = 468)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: 0.837 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 20 (n = 22)
Percentile in group and year: 18 (n = 358)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 1.025 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 67 (n = 26)
Percentile in group and year: 63 (n = 465)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: 0.024 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 71 (n = 26)
Percentile in group and year: 78 (n = 494)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 1.144 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 56 (n = 26)
Percentile in group and year: 77 (n = 481)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: 8.8 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 18.4 days)
Percentile in sector and year: 83 (n = 26)
Percentile in group and year: 78 (n = 439)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 29.61 p.p.
Alert by the study 1 rule: yes
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 75 (n = 26)
Percentile in group and year: 85 (n = 435)
The company's own history: fewer than 5 previous fiscal years with a value

Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: 0.024 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 67 (n = 26)
Percentile in group and year: 77 (n = 487)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

## Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: −0.685 (index)
Components: X1 = −0.131; X2 = −0.021; X3 = 0.018; X4 = 0.117 (indices)
Percentile in sector and year: 30 (n = 28)
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

## Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 1/9 (5 computable)
Signals:
- Return on assets (ROA): 1
- Operating cash flow on assets (CFO): 0
- Change in ROA: —
- Cash above earnings (ACCRUAL): 0
- Change in leverage: —
- Change in current ratio: 0
- No equity issuance (EQ_OFFER): —
- Change in gross margin: 0
- Change in asset turnover: —
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

## Changes

Highlighted in the following year's comparative: none.
Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets of the version used. The other differences (reclassifications between lines and smaller differences) stay in the full list.
Full list: 56 accounts above the R$1,000 tolerance (group a, totals and key lines: 5; group b, reclassifications: 51), in the annex at the end.

## What was known on the date

- FCA 2010, v1: 2010-07-10
- DFP 2010, v1: 2011-04-04
- DFP 2010, v2: 2011-04-04

- DFP 2011, v1: 2012-03-31
- DFP 2011, v2: 2012-04-02
- DFP 2011, v3: 2012-04-09

DFP and FCA filed with the CVM by the reference date; ITR comes in a later phase.
Material facts and market notices filed by the date (IPE open data, since 2010): 4 material fact documents and 40 market notice documents.
Material facts filed in the 24 months before the date (from 2010-05-01 to 2012-04-30):
- 2012-01-20: subject omitted (possible person's name); see the document at the CVM
- 2011-06-29: Migração para o Novo Mercado
- 2011-02-14: subject omitted (possible person's name); see the document at the CVM
- 2010-08-25: subject omitted (possible person's name); see the document at the CVM

Older material facts, the market notices and the link to each document are in the JSON and on screen.
Material facts and market notices filed with the CVM by the date, from the IPE open data (since 2010): filing date, category, subject (as filed, in Portuguese) and link. The content of the documents is not downloaded, read or summarized, and the list generates no alert, score or ranking. The counts are of documents: versions, resubmissions and English versions count separately; in the list, versions of the same document and documents with the same date, category and subject appear on a single line, with a link to each one.

## Data quality

53 applicable checks; 0 failed.

## Auditor's report and auditor (factual section; no alert)

Excerpt checked literally: "Em nossa opinião, as demonstrações contábeis individuais acima referidas apresentam adequadamente, em todos os aspectos relevantes, a posição patrimonial e financeira da MUNDIAL S.A. - PRODUTOS DE CONSUMO em 31 de dezembro de 2010, o desempenho de suas operações e os seus fluxos de caixa para o exercício findo naquela data, de acordo com as práticas contábeis adotadas no Brasil."
Auditor in the 2010 FCA: Directa Auditores.
Auditor in the 2011 FCA: Directa Auditores.
Auditor changes: none in the FCAs filed by the date.
Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

## Document excerpts (copied; no alert)

Excerpts copied from the documents, with the page, and the Reference Form table. It is not a signal family: it generates no alert, score or ranking. Absence of mention does not prove absence.
How the reader was tested: 20 held-out cases, a single measurement with the frozen reader (c99c953): precision 0.962, coverage 0.962, literal check on 132 of 132 excerpts, no natural-person name (docs/leitor_medida.json).

Explanatory notes · Related parties
Note: 12. PARTES RELACIONADAS (pages 16 to 17)
- [parent company] “Contas a Ativo por mútuo Passivo por Compra de Venda de Receitas Controladora receber por e conta corrente mútuo e conta produtos e produtos e financeiras vendas corrente serviços serviços Hercules S.A - Fábrica de” (p. 16, original excerpt in Portuguese)
- [associates] “O saldo de R$ 272.469 em 31 de dezembro de 2010 com a coligada Hercules S.A. teve origem na operação de” (p. 17, original excerpt in Portuguese)

- [subsidiaries] “compra em 1988 das unidades de talheres, baixelas e cutelaria (Fabricas 2 e 8) da então controlada Eberle S.A,” (p. 17, original excerpt in Portuguese)
- [subsidiaries] “As demais transações com as empresas controladas referem-se a transferências de numerários, efetuados com a finalidade de financiamento das operações das controladas.” (p. 17, original excerpt in Portuguese)

Fields of item 18 of CPC 05 (R1)
- [balances] “a) Ativo e passivo circulante e transações” (p. 16, original excerpt in Portuguese)
- [amount of the transactions; balances] “Contas a Ativo por mútuo Passivo por Compra de Venda de Receitas Controladora receber por e conta corrente mútuo e conta produtos e produtos e financeiras vendas corrente serviços serviços Hercules S.A - Fábrica de” (p. 16, original excerpt in Portuguese)
- [balances] “Saldo em 31/12/10 6.623 279.605 10.149 - 7.973 30.627” (p. 16, original excerpt in Portuguese)
- [balances] “Saldo em 31/12/09 3.776 252.748 8.831 - 8.543 23.274” (p. 16, original excerpt in Portuguese)
- [balances] “Saldo em 01/01/09 3.576 234.270 8.344 2 5.375 26.110” (p. 16, original excerpt in Portuguese)
- [amount of the transactions; balances] “Saldo ativo por Receitas Consolidado mútuo e conta financeiras corrente Hercules S.A - Fábrica de” (p. 16, original excerpt in Portuguese)
- [balances] “Saldo em 31/12/10 275.436 30.293” (p. 16, original excerpt in Portuguese)
- [balances] “Saldo em 31/12/09 246.377 23.043” (p. 16, original excerpt in Portuguese)
- [balances] “Saldo em 01/01/09 226.824 25.364” (p. 16, original excerpt in Portuguese)
- [balances] “O saldo de R$ 272.469 em 31 de dezembro de 2010 com a coligada Hercules S.A. teve origem na operação de” (p. 17, original excerpt in Portuguese)
- [amount of the transactions; balances] “compra em 1988 das unidades de talheres, baixelas e cutelaria (Fabricas 2 e 8) da então controlada Eberle S.A,” (p. 17, original excerpt in Portuguese)
- [amount of the transactions; balances] “hoje, sucedida pela Mundial S.A. A empresa Hercules S/A efetuou pagamentos em 2010 no montante de R$ 1.300, com perspectiva de crescimento considerável no próximo exercício face ao crescimento expressivo das vendas de produtos licenciados.” (p. 17, original excerpt in Portuguese)

Explanatory notes · Reverse factoring and forfaiting

The reader found no mention of reverse factoring (risco sacado), forfaiting or supplier agreements in the notes of this DFP. Absence of mention does not prove absence of the operation.

Auditor's report

Type of opinion: unqualified
- [emphasis of matter] “Ênfase Conforme evidenciado na Nota Explicativa nº 1(b), a Companhia ainda apresenta uma situação patrimonial e financeira que requer forte gestão administrativa, além disso, possui créditos a receber de empresa relacionada no valor de R$ 279.605 mil, demonstrado no realizável a longo prazo, cuja recuperação é incerta, tendo em vista a situação patrimonial e financeira daquela empresa, que apresenta capital de giro negativo e passivo a descoberto (patrimônio líquido negativo). Estes fatores são indicativos de que a continuidade normal das operações da Companhia depende do êxito das medidas relacionadas pela Administração na referida Nota Explicativa. As demonstrações contábeis não incluem quaisquer ajustes decorrentes dessas incertezas e foram elaboradas de acordo com as práticas contábeis adotadas no Brasil aplicáveis a empresas em regime normal de operações. Conforme mencionado na Nota Explicativa nº 13, a Companhia possui créditos tributários adquiridos de terceiros, reconhecidos no ativo não circulante, para os quais o reconhecimento definitivo, no entanto, depende de decisão judicial em processo que está em andamento.” (cut at the end of a sentence, original excerpt in Portuguese)
- [other matters] “Outros Assuntos Demonstrações do valor adicionado Examinamos, também, as demonstrações individual e consolidada do valor adicionado (DVA), referentes ao exercício findo em 31 de dezembro de 2010, cuja apresentação é requerida pela legislação societária

brasileira para companhias abertas, e como informação suplementar pelas IFRS que não requerem a apresentação da DVA. Essas demonstrações foram submetidas aos mesmos procedimentos de auditoria descritos anteriormente e, em nossa opinião, estão adequadamente apresentadas, em todos os seus aspectos relevantes, em relação às demonstrações contábeis tomadas em conjunto. Auditoria dos valores correspondentes ao exercício anterior Os valores correspondentes ao exercício findo em 31 de dezembro de 2009, apresentados para fins de comparação, foram anteriormente por nós auditados de acordo com as normas de auditoria vigentes por ocasião da emissão do relatório em 14.05.2010, que conteve as mesmas ênfases acima mencionadas." (cut at the end of a sentence, original excerpt in Portuguese)

Reference Form
Form of 2011, version 4, filed on 2012-03-13.
The Reference Form lists no related-party transactions in the table. It is the only version available in the open data; earlier versions are not there.

Limits of this section:
- In real documents, about 22% of the excerpts come out omitted by the name detector, including names of companies outside the registry.
- In tables, header lines (years, column titles) may come out as excerpts; the held-out cases had no tables with multi-line headers.

## How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

## Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

## Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 38 DFP lines, each with document, version, filing date, account and location.

## Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a F1); leitor-f1-1 (commit c99c953). The generation date is in the demo manifest (demo/manifesto.json).

## Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.
Name in today's registry: MUNDIAL S.A - PRODUTOS DE CONSUMO.

## Annex: full list of differences in the following year's comparative (R$ million)

(a) DRE 3.11 Lucro/Prejuízo Consolidado do Período: 10.9 in the version used, 13.0 in the

comparative (difference of 2.0)
(a) BPP 2.01 Passivo Circulante: 307.0 in the version used, 308.7 in the comparative (difference of 1.7)
(a) BPP 2.02 Passivo Não Circulante: 480.9 in the version used, 479.2 in the comparative (difference of −1.7)
(a) DFC (indirect method) 6.01 Caixa Líquido Atividades Operacionais: −10.1 in the version used, −8.4 in the comparative (difference of 1.7)
(a) BPP 2.03 Patrimônio Líquido Consolidado: 92.0 in the version used, 92.0 in the comparative (difference of −0.0)
(b) BPP 2.02.02.02.04 Impostos e Contribuições Sociais: 380.7 in the version used, 0.0 in the comparative (difference of −380.7)
(b) DFC (indirect method) 6.01.01.14 Imp. de Renda e Contr. Social Diferido - Passivo: 24.2 in the version used, −53.1 in the comparative (difference of −77.4)
(b) DFC (indirect method) 6.01.01.13 Atualização de Impostos e Contribuições: −53.1 in the version used, 0.0 in the comparative (difference of 53.1)
(b) BPP 2.01.02.01 Fornecedores Nacionais: 0.0 in the version used, 44.5 in the comparative (difference of 44.5)
(b) BPP 2.01.01 Obrigações Sociais e Trabalhistas: 75.5 in the version used, 31.8 in the comparative (difference of −43.8)
(b) BPP 2.01.01.01 Obrigações Sociais: 64.2 in the version used, 20.4 in the comparative (difference of −43.8)
(b) BPP 2.01.03 Obrigações Fiscais: 0.0 in the version used, 43.8 in the comparative (difference of 43.8)
(b) BPP 2.01.03.01 Obrigações Fiscais Federais: 0.0 in the version used, 23.8 in the comparative (difference of 23.8)
(b) DFC (indirect method) 6.01.01.10 Depreciação, Amortização e Exaustão: 13.6 in the version used, 35.2 in the comparative (difference of 21.6)
(b) BPP 2.01.03.02 Obrigações Fiscais Estaduais: 0.0 in the version used, 19.8 in the comparative (difference of 19.8)
(b) DFC (indirect method) 6.02.03 Estorno Reserva de Reavaliação: 0.0 in the version used, −11.8 in the comparative (difference of −11.8)
(b) DFC (indirect method) 6.02.04 Intangiveis: −11.8 in the version used, 0.0 in the comparative (difference of 11.8)
(b) DFC (indirect method) 6.01.01.15 Resultado não Operacional: 35.2 in the version used, 24.2 in the comparative (difference of −11.0)
(b) DFC (indirect method) 6.01.01.11 Provisão para contingencias: 3.7 in the version used, 13.6 in the comparative (difference of 10.0)
(b) BPA 1.01.02.01 Aplicações Financeiras Avaliadas a Valor Justo: 5.1 in the version used, 0.0 in the comparative (difference of −5.1)
(b) BPA 1.01.02.01.02 Títulos Disponíveis para Venda: 5.1 in the version used, 0.0 in the comparative (difference of −5.1)
(b) BPA 1.01.02.02 Aplicações Financeiras Avaliadas ao Custo Amortizado: 0.0 in the version used, 5.1 in the comparative (difference of 5.1)
(b) BPA 1.01.02.02.01 Títulos Mantidos até o Vencimento: 0.0 in the version used, 5.1 in the comparative (difference of 5.1)
(b) DFC (indirect method) 6.01.02.11 Provisões para Contingencias: −3.5 in the version used, 0.0 in the comparative (difference of 3.5)
(b) DFC (indirect method) 6.01.01.01 Resultado Liquido do Exercício: 10.9 in the version used, 12.9 in the comparative (difference of 2.0)
(b) DRE 3.06 Resultado Financeiro: 16.2 in the version used, 18.3 in the comparative (difference of 2.0)
(b) DRE 3.06.02 Despesas Financeiras: −74.2 in the version used, −72.2 in the comparative (difference of 2.0)
(b) DRE 3.07 Resultado Antes dos Tributos sobre o Lucro: 31.7 in the version used, 33.7 in the comparative (difference of 2.0)
(b) DRE 3.09 Resultado Líquido das Operações Continuadas: 10.9 in the version used, 13.0 in the

comparative (difference of 2.0)
(b) DRE 3.11.01 Atribuído a Sócios da Empresa Controladora: 10.9 in the version used, 12.9 in the comparative (difference of 2.0)
(b) DFC (indirect method) 6.01.01.18 Provisões Diversas Passivos: 0.2 in the version used, −1.8 in the comparative (difference of −2.0)
(b) DFC (indirect method) 6.03.03 Juros de Capital Proprio Propostos: 1.7 in the version used, −0.1 in the comparative (difference of −1.8)
(b) BPP 2.01.05 Outras Obrigações: 18.3 in the version used, 20.0 in the comparative (difference of 1.7)
(b) BPP 2.01.05.02.01 Dividendos e JCP a Pagar: 0.0 in the version used, 1.7 in the comparative (difference of 1.7)
(b) BPP 2.02.02 Outras Obrigações: 383.1 in the version used, 381.4 in the comparative (difference of −1.7)
(b) BPP 2.02.02.02 Outros: 383.1 in the version used, 381.4 in the comparative (difference of −1.7)
(b) BPP 2.02.02.02.05 Juros de Capital Próprio: 1.7 in the version used, 0.0 in the comparative (difference of −1.7)
(b) DFC (indirect method) 6.03 Caixa Líquido Atividades de Financiamento: 19.1 in the version used, 17.4 in the comparative (difference of −1.7)
(b) DFC (indirect method) 6.01.02 Variações nos Ativos e Passivos: −9.4 in the version used, −7.7 in the comparative (difference of 1.7)
(b) BPP 2.01.05.02 Outros: 18.3 in the version used, 19.7 in the comparative (difference of 1.4)
(b) DFC (indirect method) 6.01.02.06 Imposto de Renda e Contribuição Social Diferidos: 0.0 in the version used, −1.1 in the comparative (difference of −1.1)
(b) BPP 2.03.05 Lucros/Prejuízos Acumulados: −45.8 in the version used, −44.8 in the comparative (difference of 1.0)
(b) BPP 2.03.01 Capital Social Realizado: 29.7 in the version used, 28.8 in the comparative (difference of −0.9)
(b) BPA 1.02.01.01 Aplicações Financeiras Avaliadas a Valor Justo: 0.7 in the version used, 0.0 in the comparative (difference of −0.7)
(b) BPA 1.02.01.01.02 Títulos Disponíveis para Venda: 0.7 in the version used, 0.0 in the comparative (difference of −0.7)
(b) BPA 1.02.01.02 Aplicações Financeiras Avaliadas ao Custo Amortizado: 0.0 in the version used, 0.7 in the comparative (difference of 0.7)
(b) BPA 1.02.01.02.01 Títulos Mantidos até o Vencimento: 0.0 in the version used, 0.7 in the comparative (difference of 0.7)
(b) DFC (indirect method) 6.01.02.13 Outras Contas a Pagar: 0.4 in the version used, 0.0 in the comparative (difference of −0.4)
(b) BPP 2.01.05.01 Passivos com Partes Relacionadas: 0.0 in the version used, 0.3 in the comparative (difference of 0.3)
(b) BPP 2.01.05.02.04 Outras Contas a Pagar: 18.3 in the version used, 17.9 in the comparative (difference of −0.3)
(b) BPP 2.03.04 Reservas de Lucros: 27.3 in the version used, 26.9 in the comparative (difference of −0.3)
(b) BPP 2.03.04.10 Reserva Orçamento de Capital: 24.1 in the version used, 23.7 in the comparative (difference of −0.3)
(b) BPP 2.03.06 Ajustes de Avaliação Patrimonial: −1.5 in the version used, −1.3 in the comparative (difference of 0.2)
(b) BPP 2.01.03.03 Obrigações Fiscais Municipais: 0.0 in the version used, 0.2 in the comparative (difference of 0.2)
(b) BPP 2.01.02 Fornecedores: 44.5 in the version used, 44.5 in the comparative (difference of 0.0)
(b) DFC (indirect method) 6.01.01.12 Variação Cambial de Fornecedores: 0.0 in the version used, 0.0 in the comparative (difference of −0.0)